

4.5.1 mapping read values on bit string

Q: $[0, 10] \rightarrow 3 \text{ bit}$

$$\Gamma(a_1, \dots, a_L) = x + \frac{y-x}{2^L-1} \left(\sum_{j=0}^{L-1} a_{L-j} \cdot 2^j \right) \in [x, y]$$

Solution

① $x = 0$

$y = 10$

$L = 3$

$$\begin{aligned} \Gamma(a_1, a_2, a_3) &= 0 + \frac{10-0}{2^3-1} \left(\sum_{j=0}^2 a_{3-j} \cdot 2^j \right) \\ &= \frac{10}{7} (a_3 + a_2 \cdot 2 + a_1 \cdot 2^2) \\ &\quad \text{equal to 10 进制} \end{aligned}$$

3 bit	10 进制	Mapping to $[0, 10]$
000	0	0
001	1	1.43
010	2	2.86
011	3	4.29
100	4	5.71
101	5	7.14
110	6	8.57
111	7	10

$$B = \text{round} \left[\frac{z - x}{y - x} (2^L - 1) \right]$$

$$z = 1.1 \quad [x, y] = [-10, 10] \quad \rightarrow 6 \text{ bits}$$

$$B = \text{round} \left[\frac{1.1 - (-10)}{10 - (-10)} (2^6 - 1) \right]$$

$$= \text{round} \left[\frac{1.1 + 10}{20} \times 63 \right]$$

$$= \text{round}(34.965)$$

$$= 35$$

$$\begin{array}{r} 1 \overline{) 35} \\ 1 \overline{) 17} \\ 0 \overline{) 8} \\ 0 \overline{) 4} \\ 0 \overline{) 2} \\ 1 \end{array}$$

$$[00011]$$